Nithin Shet N 26110690025

Comparing Permissions on Regular Files and Directories

1.Compare Permissions on Regular Files and Directories.

Default Permission	User	Group	Others
File	Read, write	Read	read
Directory	Read, write, execute	Write, execute	execute

```
[nithinshetn@Nithins-MacBook-Air linuxTheory % ls -l sample.txt                    ]
 -rw-r--r--  1 nithinshetn  staff  0 22 Aug 22:47 sample.txt
[nithinshetn@Nithins-MacBook-Air linuxTheory % ls -ld sample                       ]
 drwxr-xr-x  2 nithinshetn  staff  64 22 Aug 22:47 sample
 nithinshetn@Nithins-MacBook-Air linuxTheory %
```

2. Apply each of the following permissions using chmod. Perform the specified operations and record your observations.

- Read: cat sample.txt
- Write: echo “writing” > sample.txt
- Change permission: chmod 000 sample.txt

Permission	Regular File (sample.txt) – Test	Observation
400 (r--)	Try to read and modify the file	• read - allowed • write - when tried to modify shows ‘readonly’
200 (-w-)	Try to read and modify the file	• read - not allowed • write - possible using echo >sample.txt and using vi not able to write
100 (--x)	Try to read and execute the file	• read – denied • write – denied (allowed only for executable files)
500 (r-x)	Try to read and modify the file	• read- allowed • write- Permission denied • execute- allowed(but invalid)
600 (rw-)	Try to read and modify the file	• read- allowed

		• write- allowed • execute- not allowed
700 (rwx)	Try to read, modify, and execute the file	• read and write allowed

- **DIRECTORY**

Permission	Directory (sample_dir) – Test	Observation
400 (r--)	Try to list and enter the directory	• dr-------- 2 sa sa 4096 Aug 22 10:23 sample_dir • list: ls: cannot access 'sample_dir/file1.txt': Permission denied file1.txt • enter: denied
200 (-w-)	Try to list, enter, and create a file	• list: denied • enter: denied • create: denied
100 (--x)	Try to list and enter the directory	• list: denied • enter: denied
500 (r-x)	Try to list and enter the directory	• list: denied • enter: allowed and can access the files
600 (rw-)	Try to list, enter, and create a file	• list: denied • enter: denied • create: denied ls -l sample_dir ls: cannot access 'sample_dir/file1.txt': Permission denied total 0 -????????? ? ? ? ? ? file1.txt
700 (rwx)	Try to list, enter, create, and delete files	All allowed

Questions

- What does read (r) permission allow you to do with a regular file?

  Allows read/view content of the file.

- What does read (r) permission allow you to do with a directory?

  List the names of the files and directories.

- What does write (w) permission mean for a regular file?

  Modify, overwrite, or write data to the file.

- What does write (w) permission mean for a directory?

  Create, delete, and rename files or directories inside it.

- What does execute (x) permission mean for a regular file?

  Allows the file to be executed as a program/script, provided it is a valid executable or script.

- Why is execute (x) permission important for accessing a directory?

  Enter/traverse the directory and access items inside it.

- Which permission combination allows the owner to fully access a regular file

  rwx (700)

- Which permission combination allows the owner to fully access a directory??

  rwx (700)

- Based on your observations, explain why the same r, w, and x permissions behave differently for a regular file and a directory.

  File contains data and directory contains entries like filea , directories. Therefore, the permissions have different meanings.

  - File: **r** means reading contents, **w** means modifying contents and **x** means execute/run scripts or files.
  - Directory: **r** means listing entries, **w** means creating/deleting entries, and **x** means entering or traversing the directory.